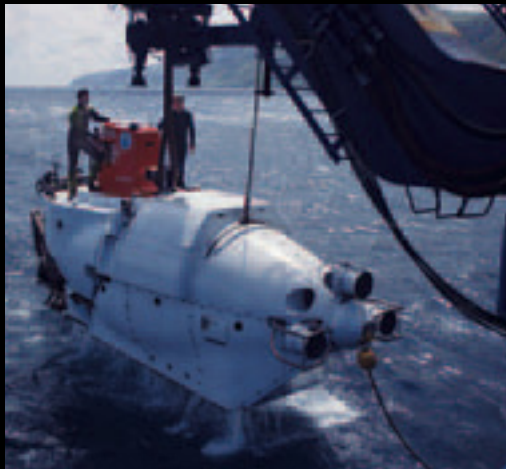


EXPLORING THE OCEAN DEPTHS

For centuries, people have been exploring shallow seas using simple diving bells. However, the crushing pressures at greater depths made deep-ocean exploration impossible, until the development of deep-water submersibles.

The first of these were little more than pressure-proof metal spheres, equipped with weights and floats for sinking and rising. *Alvin*, built in 1962, was the first of a new generation of manned craft that could be easily maneuvered underwater to gather samples and record images of the deep ocean floor.



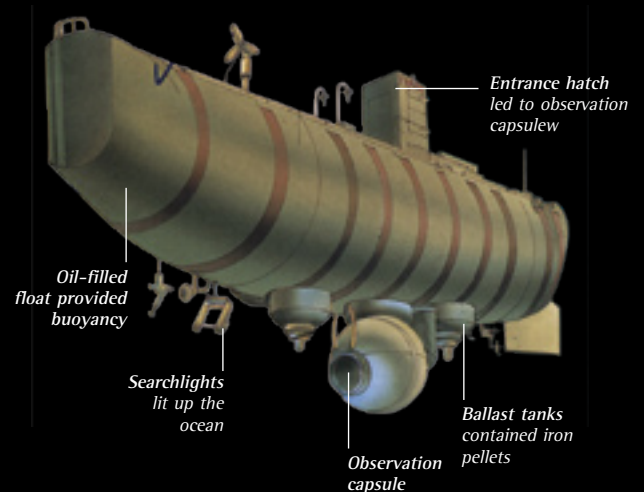
◀ ALVIN

Owned by the US Navy, but operated by the Woods Hole Oceanographic Institution in New England, *Alvin* can carry two scientists and a pilot to a maximum depth of 15,000 ft (4,500 m). Since its first dive in 1964, it has made more than 4,000 descents, discovering deep-sea hydrothermal vents in 1977 and making the first manned survey of the *Titanic*.



◀ DIVING BELL

Diving bells were being used for salvaging wrecks as early as 1690. These were often built of wood, made watertight above, but with an opening at the bottom, and suspended from a mother ship by a cable. As the bell was lowered to the seabed, air pressure inside it stopped the water level from rising inside. A diver could also work on the seabed wearing a helmet to which air was piped from the diving bell.



▲ THE BATHYSCAPHE

In 1960, Jacques Piccard (born 1922) and Don Walsh (born 1931) descended into the deepest part of the Pacific in the bathyscaphe *Trieste*. This was basically a metal sphere suspended from a huge oil-filled float, and weighed down with iron ballast. After sinking for several hours, they spent 20 minutes on the ocean floor before releasing the ballast to return to the surface. But they saw a fish there, proving that life exists in extreme depths.

◀ MANNED SUBMERSIBLES

At a depth of about 13,000 ft (4,000 m), a dive to the ocean floor takes at least two hours, with the scientists and pilot crammed into a small and uncomfortable but very strong metal sphere. They leave the outside lights off to save power, but switch them on when they reach the bottom so that they can see, explore, gather samples, and send video images back to the mother ship.

